

Abdelrhman Saeed

AI Engineer

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EDUCATION

Arab Academy for Science, Technology, and Maritime

Bachelors in Artificial Intelligence (GPA: 3.2/4.00 – “Very Good” Academic Standing)

El-Alamein, Egypt

Expected Aug 2025

- Concentration: Robotics and Intelligent Systems
- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Machine Learning, Deep Learning, Reinforcement Learning, Computer Vision, NLP, HPC, Cloud Computing, Image Processing, Robotics, Kinematics, Path Planning, Localization, Swarm Algorithms

EXPERIENCE

Machine Learning Course - ITI

Trainee

Alexandria, Egypt

Aug 2024 – Sep 2024

- Completed a 60-hour machine learning course covering supervised and unsupervised learning through practical projects.

ICPC Competitive Programmer

Team Leader & Contestant

Alexandria, Egypt

Jun 2023 – Jun 2024

- Led college team in national ICPC contests.
- Finalist in ECPC and ACPC. Ranked 1st in college in both ECPC finals. Achieved first solve in 2024.

RoboCup@Home EDU Workshop

AI Robotics Specialist

Alexandria, Egypt

Feb 2023 – Mar 2023

- Programmed the Jupiter robot with C++ and ROS. Applied vision techniques to enable interaction.

Competitive Programming

Active Problem Solver

Online

Jun 2021 – Present

- Codeforces: Achieved Expert-level rating of **1616** (max), solved over **1700** rated problems – placing in the 1600–1899 “Expert” tier among 1.7M+ users

PROJECTS

B.E.M.O – Multimodal Assistant Robot — Python, LangChain, Tavily, OpenCV, FastAPI

Jul 2024 – Present

- Developed “Bemo,” a desktop assistant robot leveraging LangChain and Tavily for multimodal understanding.
- Used OpenCV and NLP libraries to process user inputs and automate task execution with 95% reliability.
- Integrated Google Gemini LLM to enhance dialogue reasoning.
- Built modular FastAPI services to orchestrate actions and AI backends.
- Logged user interactions and model responses to improve contextual memory and fallback performance.

Face Analysis & Recognition System — Python, C++, MediaPipe, LFW

Jun 2025

- Benchmarked 11 face-analysis models – achieved 99–100% detection accuracy and 98.6% recognition with a 0.16s/image pipeline.
- Engineered optimized C++ recognition using MediaPipe and TensorFlow Lite for edge inference on Raspberry Pi.
- Compared model accuracy using LFW embeddings and visualized outputs to evaluate landmark precision.

Lip Reading System — Python, 3D-CNN, OpenCV

Sep 2023 – Oct 2023

- Created a system using 3D CNNs to transcribe silent video lip movements into text.
- Achieved strong word-level accuracy on curated dataset.
- Implemented preprocessing with face tracking, mouth cropping, and temporal alignment for video slices.

Mimic-NAO: Human-to-Robot Pose Control — NAOqi, Inverse Kinematics

Jan 2023 – Feb 2023

- Engineered a real-time robot mimicry system using MediaPipe and NAOqi with 96%+ pose accuracy.
- Implemented both analytical and Jacobian-based inverse kinematics for smooth full-body mapping.

Cubot: Rubik’s Cube Solver — Python, OpenCV, Arduino

Sep 2022 – Jan 2023

- Designed a Rubik’s Cube solving robot using color detection, the Kociemba algorithm, and Arduino-based actuation.
- Optimized servo actuation timing and color calibration for consistent sub-30 second solve time.

CERTIFICATES & ACHIEVEMENTS

Qualified to ACPC Regionals (2024); won **First Solve Award** at ECPC 2024; ranked in **Top 50** Arab teams.

Placed **Top 10** in ECPC Qualification Day (2023, 2024); **1st Place** in ECPC College Qualifiers both years.

RoboCup@Home EDU Workshop (Feb–Mar 2023): Vision, robotics, and autonomous interaction.

DEBI Robotics Challenge, Ministry of Communications (Jul–Aug 2022): TurtleBot3 navigation and obstacle avoidance.

TECHNICAL SKILLS

Languages: Python; C/C++; SQL; Dart; JavaScript; HTML/CSS

Frameworks/Tools: TensorFlow; OpenCV; ROS; FastAPI; Docker; Git; MATLAB

Platforms: Linux; Windows; Arduino; AWS

IDEs: VS Code; CLion; Eclipse

LEADERSHIP & VOLUNTEERING

Tech Volunteer, RoboCup Egypt Nationals – Rescue Maze League (2025): Evaluated teams based on mission success, strategy, and robot performance.

Founder & Coach (Certified), Competitive Programming Communities – AAST Alamein & Assiut: Created CP groups and led training in algorithms and contests.

Team Member, Sky Experts Drones Software Team – Jan 2024 – Sep 2024: Developed quadcopter control algorithms using ROS, focused on CV-based mission planning.